

September 2026 Exam

EC400

Probability and Statistical Inference

Sample Exam 2

Suitable for all candidates

Instructions to candidates

This paper contains **SIX** questions. Answer **ANY FOUR**.

Each question carries 25 marks. Four attempted questions carry 100 marks in total.

Show the reasoning and calculations supporting every answer.

If more than four questions are attempted, the four answers awarded the highest marks will count toward the examination mark.

State clearly any additional assumption that you use.

Time Allowed	Reading Time: 15 minutes
	Writing Time: 2 hours

You are supplied with:	<i>Standard Normal and Student's t tables</i>
-------------------------------	--

You may also use:	<i>No additional materials</i>
--------------------------	--------------------------------

Calculators:	<i>Calculators are not allowed in this examination</i>
---------------------	--

If you need an additional assumption in order to proceed, state it clearly before using it.

Question 1

Employees in one branch of a firm may choose whether to attend an optional coaching programme. Let $D = 1$ denote attendance, let $Y(d)$ be the performance score that would be observed under programme status d , and let $Y = DY(1) + (1 - D)Y(0)$. Assume all relevant expectations exist, and write $p = \Pr(D = 1)$, where $0 < p < 1$. Define $\beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$. **[25 marks]**

- (a) Define the individual treatment effect, the average treatment effect and the average treatment effect on the treated. Explain why only one potential outcome is observed for each employee. **[5 marks]**
- (b) Derive the decomposition of β into the ATET and a selection-bias term. Explain what each term represents. **[5 marks]**
- (c) Use the law of iterated expectations to prove that

$$\text{ATE} = p \text{ ATET} + (1 - p)\mathbb{E}[Y(1) - Y(0) \mid D = 0].$$

Explain separately why the ATE may differ from the ATET and why β may differ from the ATET. **[5 marks]**

- (d) Suppose instead that programme status were randomly assigned within this branch. State the resulting potential-outcome independence condition and its implications for β , the ATET and the ATE. Would this alone identify the ATE for every branch of the firm? **[5 marks]**
- (e) In an observational sample, let n_1 and n_0 be the numbers of attendees and non-attendees, and let $\bar{Y}_1$ and $\bar{Y}_0$ be their sample means. State a natural estimator of β . Distinguish an estimator, an estimate and a parameter. Under random sampling within each group, show why increasing both group sizes improves estimation of β but cannot by itself remove selection bias. **[5 marks]**

Question 2

The joint density of continuous random variables (X, Y) is

$$f_{X,Y}(x, y) = x + y, \quad 0 < x < 1, \quad 0 < y < 1,$$

and is zero otherwise.

[25 marks]

- (a) Derive the two marginal densities and the conditional density of X given $Y = y$. State all relevant supports. **[6 marks]**
- (b) Calculate $\mathbb{E}[X \mid Y = y]$. Then calculate $\mathbb{E}[X]$ directly from f_X and confirm that averaging the conditional mean recovers the same result. **[6 marks]**
- (c) Calculate $\mathbb{E}[XY]$ and $\text{Cov}(X, Y)$. Are X and Y independent? **[7 marks]**
- (d) Let $W = X + Y$. Calculate $\mathbb{E}[W]$ and $\text{Var}(W)$. **[6 marks]**

Question 3

Let $U \sim U(0, 1)$, let $\lambda > 0$, and define

$$T = -\frac{1}{\lambda} \log(1 - U).$$

This transformation is strictly increasing. Let $T_1, T_2, \dots$ be independent copies of T , and let $S_m = \sum_{j=1}^m T_j$, where m is a positive integer. **[25 marks]**

Reference formulas. You may use the following, with $b, \nu > 0$ and positive integer a :

- If $W \sim \text{Exp}(b)$, then $\mathbb{E}[W] = 1/b$ and $\text{Var}(W) = 1/b^2$.
- A $\text{Gamma}(a, b)$ variable (shape a , rate b) has density

$$f_G(s) = \frac{b^a}{(a-1)!} s^{a-1} e^{-bs}, \quad s > 0,$$

and zero otherwise.

- If $N \sim \text{Poisson}(\nu)$, then

$$\Pr(N = j) = e^{-\nu} \frac{\nu^j}{j!}, \quad j = 0, 1, \dots, \quad \mathbb{E}[N] = \text{Var}(N) = \nu.$$

- (a) State the support of T and find the inverse transformation. **[5 marks]**
- (b) Use the CDF method to derive the CDF and density of T over the whole real line. Name the distribution and give its parameter. **[5 marks]**
- (c) Derive the tail probability $\Pr(T > t)$ and median of T . Give its expectation and variance, and explain how increasing λ changes the distribution. **[5 marks]**
- (d) State the distribution of S_m , including its parameterisation. Write its density and give its expectation and variance. **[5 marks]**
- (e) Interpret the T_j as successive waiting times between arrivals and, for $t > 0$, let $N(t)$ be the number of arrivals by time t . State the distribution of $N(t)$, express $\Pr(S_m \leq t)$ in terms of $N(t)$, and derive its probability. Also give $\mathbb{E}[N(t)]$ and $\text{Var}(N(t))$. **[5 marks]**

Question 4

For four observations, let

$$z = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 6 \end{pmatrix}.$$

An intercept is included in the linear regression of y on z . You may use the familiar OLS formula $\hat{\beta} = (X'X)^{-1}X'y$; no derivation of this formula is required. **[25 marks]**

- (a) Write the design matrix X and calculate $X'X$. **[4 marks]**
- (b) Calculate $X'y$ and $(X'X)^{-1}$, and hence obtain the OLS intercept and slope. **[5 marks]**

- (c) Calculate the fitted values and residuals. Verify $X'\hat{u} = 0$ and explain why a fitted residual is not the same object as the unobserved population error ε_i . **[5 marks]**
- (d) Suppose the conditional covariance matrix of the population errors is

$$\Omega = \text{Var}(\varepsilon | X) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Describe the variance and dependence pattern. Calculate $X'\Omega X$ and use

$$\text{Var}(\hat{\beta} | X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$$

to obtain the exact conditional covariance matrix of $\hat{\beta}$ and the two corresponding standard errors. In a large sample with this covariance pattern and suitable technical conditions, which robust standard-error method is appropriate, and why does it leave $\hat{\beta}$ unchanged? **[7 marks]**

- (e) Suppose z_i records self-selected training hours. Interpret the estimated slope and explain why it is not automatically a causal effect. What does $\mathbb{E}[\varepsilon_i | z_i] = 0$ imply, and what kind of design would strengthen a causal interpretation? **[4 marks]**

Question 5

For an estimator $\hat{\beta}_n$, suppose that, for every sample size n ,

$$\hat{\beta}_n \sim N\left(\beta, \frac{\sigma^2}{n}\right),$$

where $\sigma > 0$ is known. Before observing the data, a researcher chooses to test $H_0 : \beta \leq 0$ against $H_1 : \beta > 0$ at significance level α . Write $z_{1-\alpha}$ for the corresponding standard-Normal quantile and let

$$Z_n = \frac{\sqrt{n}\hat{\beta}_n}{\sigma}.$$

The null includes zero and negative effects. To control the chance of rejecting a true null, we must consider every value $\beta \leq 0$. The largest of these rejection probabilities is called the size of the test.

[25 marks]

- (a) Consider the rule that rejects when $Z_n > z_{1-\alpha}$. Find its rejection probability for a value $\beta \leq 0$. Show that this probability is largest at $\beta = 0$, where it equals α , so the test has size α . Explain why the direction of the test must be chosen before the estimate is observed. **[5 marks]**
- (b) Given an observed value z_{obs} , use the null value $\beta = 0$ from part (a) to obtain the right-tail p -value. Show that comparing it with α gives the same decision as comparing z_{obs} with the critical value. Give a correct frequentist interpretation of the p -value. **[5 marks]**
- (c) Consider the lower one-sided confidence set

$$C_{1-\alpha} = \left[\hat{\beta}_n - z_{1-\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right).$$

Show that $\Pr_{\beta}(\beta \in C_{1-\alpha}) = 1 - \alpha$: this is the interval's repeated-sampling coverage. Show also that the interval excludes zero exactly when the test rejects, and explain the coverage statement in words. **[5 marks]**

- (d) For a fixed alternative $\beta > 0$, derive the power of the test. Show that power increases with β and with n , and decreases with σ . What happens to power as $n \rightarrow \infty$? **[5 marks]**
- (e) Suppose we carry out M tests. A *false rejection* means rejecting a null hypothesis that is true. We want the probability of *at least one* false rejection across all M tests to be at most α . This is often called controlling the *family-wise error probability*.
The Bonferroni rule tests each null at level α/M . Use the union bound to show that the probability of at least one false rejection is at most α , even without independence.
For an exact calculation, now assume the tests are independent and all nulls are true, with each test having actual rejection probability α/M (as at $\beta = 0$ in part (a)). First find the probability of no false rejections, then the probability of at least one. Compare the latter with α .
What is the cost of the smaller per-test level in terms of power? **[5 marks]**

Question 6

Let u_t , for $t \in \mathbb{Z}$, be independent and identically distributed with $\mathbb{E}[u_t] = 0$ and $\text{Var}(u_t) = \sigma^2$, where $0 < \sigma^2 < \infty$. Consider the time series

$$X_t = \mu + u_t + u_{t-1}.$$

[25 marks]

- (a) Calculate $\mathbb{E}[X_t]$, $\text{Var}(X_t)$ and $\text{Cov}(X_t, X_{t-h})$ for $h = 1$ and for $|h| \geq 2$. **[5 marks]**
- (b) State the definition of weak stationarity in terms of first and second moments, and verify that (X_t) is weakly stationary. **[5 marks]**
- (c) For $\bar{X}_n = n^{-1} \sum_{t=1}^n X_t$, derive $\text{Var}(\bar{X}_n)$ and show that $\bar{X}_n \xrightarrow{p} \mu$. You may use the following probability bound: for any finite-variance random variable Z and any $a > 0$,

$$\Pr(|Z - \mathbb{E}[Z]| > a) \leq \frac{\text{Var}(Z)}{a^2}.$$

[5 marks]

- (d) A researcher incorrectly treats the observations as uncorrelated. Compare the resulting variance formula for $\bar{X}_n$ with the correct large- n variance. Does this understate or overstate uncertainty? Which standard-error method from Lecture 4 is designed to allow for this serial correlation? **[5 marks]**
- (e) Now consider the time-series regression $y_t = x_t' \beta + \varepsilon_t$ and the algebraic identity

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{t=1}^n x_t x_t' \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n x_t \varepsilon_t \right).$$

State probability-limit conditions sufficient for OLS consistency, and explain how moving from cross-sectional to time-series data changes the argument. **[5 marks]**

END OF PAPER

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576